

Introduction to Java

1. Low Level Language

A low level language is a language that the computer can understand directly. It is written in binary numbers (0 and 1), so it is difficult for humans to read and write.

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- * Easy for computer to understand.
- * Difficult for programmers to write.

2. High Level Language

A high level language is a language that is easy for humans to read, write, and understand. It uses English-like keywords and symbols.

- * Easy to learn and write.
- * Easy to maintain.
- * Needs a translator to convert it into machine language.

Examples: Java, Python, C++, C#

Java Basic Introduction

*Java is a high-level programming language developed by Sun Microsystems in 1995.

Code Example :

```
public class Demo {  
    public static void main(String[] args) {  
        System.out.println("Welcome to Java");  
    }  
}
```

- * Java is easy to learn.
- * Java is widely used for web, desktop, and enterprise applications.
- *Java is Both **Compiled** and **Interpreted** Language

Java is called both a Compiled Language and an Interpreted Language because it uses both a Compiler and an Interpreter during program execution.

How It Works

Step 1: **Compilation**

A compiler is a software that converts the entire source code into bytecode before execution.

The Java Compiler (javac) converts the source code (.java) into bytecode (.class).

Key Points :

- * Checks compile-time errors.
- * Converts Java code into bytecode.
- * Generates a .class file.

Demo.java



Compiler (javac)



Demo.class

Step 2: **Interpretation**

An interpreter executes the code line by line.

The JVM reads the bytecode and executes it .

Key Points :

- * Executes the program.
- * Produces output on the screen.
- * Works after compilation is completed.

Demo.class



JVM



Output

Example

```
public class Demo {  
    public static void main(String[] args) {  
        System.out.println("Hello Java");  
    }  
}
```

Compilation > javac Demo.java

Execution > java Demo

Why Java is Both?

Java first compiles source code into bytecode.

Then JVM interprets and executes the bytecode.

Therefore, Java uses both compilation and interpretation.

Difference Between Compiler and Interpreter

Compiler

Converts the entire program at once.

Execution is faster after compilation.

Errors are shown after the whole program is compiled.

Example: Java Compiler (javac)

Interpreter

Executes the program line by line.

Execution is comparatively slower.

Errors are shown one by one during execution.

Example: JVM Interpreter

In Java, both Compiler and Interpreter are used.

